I worked on Synexa, a RAG-based document assistant that allows users to upload documents and ask questions in natural language. One challenge I found interesting was deciding what “correct” should mean for an AI-generated answer.

A response can sound convincing and still be wrong. For Synexa, I decided that a correct answer should not only answer the user's question clearly; it should also be based on relevant information actually retrieved from the uploaded document. This influenced how I designed the system. Documents were split into chunks, converted into embeddings, and stored in FAISS. When a user asked a question, the system retrieved the most relevant chunks and passed them as context to the language model before generating a response.

I measured the system at multiple stages rather than judging only the final answer. I tested whether valid documents were processed successfully, large documents were chunked correctly, embeddings were stored, and relevant chunks were retrieved for matching queries. I also tested whether the generated response used the query and retrieved context appropriately. Invalid uploads and cases where no document had been uploaded were included as failure cases. All of these test cases passed in the project testing.

I also tested the system with documents of different sizes. Through this process, I learned that answer quality depended heavily on retrieval quality and on how well the documents were preprocessed and chunked. Ambiguous questions could still produce less accurate responses, which was an important limitation to recognize rather than ignore.

This project taught me that for an AI system, “correct” is not simply about producing an answer. It is about building a process where the answer is supported by relevant evidence and where the system's limitations can be tested and understood